KO LIH HAN

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WORK EXPERIENCE

Software Developer Aug 2022 – Aug 2026
Kahap Inc | Part-time

- Developed core features for a production Laravel/Vue.js feedback platform, including submissions, discussions, administrative review, notifications, and scheduled tasks.
- Managed day-to-day production operations across deployments, queue workers, scheduled jobs, database migrations, SSL renewal, and Linux server troubleshooting.
- Integrated GPT-SoVITS into backend speech workflows with pronunciation and text validation before synthesis.

Full-Stack Developer Intern Jun 2021 – Jan 2022
Pac Com Solution

- Developed and maintained features for a production Laravel/Vue.js point-of-sale system across backend logic and frontend interfaces.

RESEARCH EXPERIENCE

Interpreting Implicit Intent through Workplace Hierarchies and Context

M.S. Thesis, National Tsing Hua University | NLP, Graph Learning, Knowledge Graphs, LLMs

- Modeled how workplace context and speaker-listener roles affect the implicit intent of an utterance.
- Represented speaker and listener states as dialogue graphs, and retrieved role- and context-specific knowledge to guide Llama 3.1-8B.
- Evaluated the approach on 400 cases where only the target utterance was available; combining role and context knowledge improved semantic F1 from 0.644 to 0.682.

PROJECTS

Small-Agent QLoRA

Qwen3.5-9B, QLoRA, Tool Use, Agent Evaluation

- Built a Qwen-based agent that can search, inspect files, and execute Python to complete multi-step tasks.
- Fine-tuned the model on examples collected from failed agent runs, improving task success from 34.5% to 62.0% across 200 evaluation tasks.
- On GAIA100, accuracy improved from 9% to 18%; error analysis showed excessive searching remained the main failure mode.

Hybrid Retrieval Evaluation

BM25, Dense Retrieval, RRF, Retrieval Evaluation

- Compared lexical, dense, and hybrid retrieval for finding relevant documents from a 5,189-document collection.
- Combining BM25 and dense retrieval improved Recall@10 from 0.734 to 0.763 and MRR@20 from 0.679 to 0.742 across 64 queries.

Semantic Text-to-SQL

Qwen3.5-9B, Schema Grounding, SQL Validation, Agentic Repair

- Built a Text-to-SQL system that uses database schema information to generate, check, and repair SQL before execution.
- On 100 BIRD DEV queries, correct-query accuracy improved from 33% to 39%, fixing 6 previously incorrect cases without introducing regressions.

EDUCATION

National Tsing Hua University Jun 2026
M.S. in Information Systems & Applications | GPA 4.09/4.3

Chung Hua University Jun 2024
B.S. in Computer Science & Information Engineering | GPA 3.9/4.0

SKILLS

LLM / NLP:	RAG, hybrid retrieval, knowledge graphs, tool-using agents, PEFT / QLoRA, LLM evaluation
Machine Learning:	Transformers, graph neural networks, embedding models
Backend:	Python, FastAPI, PHP, Laravel, REST APIs, SQL, MySQL, SQLite
Tools:	Linux, Git, pytest, llama.cpp, LangGraph
Languages:	Chinese (Native), English (Fluent), Malay (Intermediate)